Toy ECDSA by hand: $y^2 = x^3 + 7$ over F_{67}, #E = 79 (prime)

Coordinates are reduced mod 67 and scalars mod 79. Generator G = (2, 22) (any point ≠ O generates, because 79 is prime). Private key d = 5, nonce k = 14. Inverses mod 67 come from extended Euclid; for example $44 \cdot 32 = 1408 = 21 \cdot 67 + 1$, so $44^{-1} \equiv 32$. Every value below is regenerated from `toy_ecdsa.py`.

1 · Point arithmetic (Double-and-Add, left to right)

operation	λ (mod 67)	$x_3 = \lambda^2 - x_1 - x_2 \pmod{67}$	$y_3 = \lambda(x_1 - x_3) - y_1 \pmod{67}$	result
Key: Q = d·G = 5·G (5 = 101_2: double, double, add)				
2·(2, 22)	$(3 \cdot 2^2)/(2 \cdot 22) = 12 \cdot 44^{-1} = 12 \cdot 32 \equiv 49$	$49^2 - 2 \cdot 2 = 2397 \equiv 52$	$49 \cdot (2 - 52) - 22 = -2472 \equiv 7$	(52, 7)
2·(52, 7)	$(3 \cdot 52^2)/(2 \cdot 7) = 5 \cdot 14^{-1} = 5 \cdot 24 \equiv 53$	$53^2 - 2 \cdot 52 = 2705 \equiv 25$	$53 \cdot (52 - 25) - 7 = 1424 \equiv 17$	(25, 17)
(25, 17) + (2, 22)	$(22 - 17)/(2 - 25) = 5 \cdot 44^{-1} = 5 \cdot 32 \equiv 26$	$26^2 - 25 - 2 = 649 \equiv 46$	$26 \cdot (25 - 46) - 17 = -563 \equiv 40$	(46, 40)
Nonce point: R = k·G = 14·G (14 = 1110_2: dbl, add, dbl, add, dbl)				
2·(2, 22)	$(3 \cdot 2^2)/(2 \cdot 22) = 12 \cdot 44^{-1} = 12 \cdot 32 \equiv 49$	$49^2 - 2 \cdot 2 = 2397 \equiv 52$	$49 \cdot (2 - 52) - 22 = -2472 \equiv 7$	(52, 7)
(52, 7) + (2, 22)	$(22 - 7)/(2 - 52) = 15 \cdot 17^{-1} = 15 \cdot 4 \equiv 60$	$60^2 - 52 - 2 = 3546 \equiv 62$	$60 \cdot (52 - 62) - 7 = -607 \equiv 63$	(62, 63)
2·(62, 63)	$(3 \cdot 62^2)/(2 \cdot 63) = 8 \cdot 59^{-1} = 8 \cdot 25 \equiv 66$	$66^2 - 2 \cdot 62 = 4232 \equiv 11$	$66 \cdot (62 - 11) - 63 = 3303 \equiv 20$	(11, 20)
(11, 20) + (2, 22)	$(22 - 20)/(2 - 11) = 2 \cdot 58^{-1} = 2 \cdot 52 \equiv 37$	$37^2 - 11 - 2 = 1356 \equiv 16$	$37 \cdot (11 - 16) - 20 = -205 \equiv 63$	(16, 63)
2·(16, 63)	$(3 \cdot 16^2)/(2 \cdot 63) = 31 \cdot 59^{-1} = 31 \cdot 25 \equiv 38$	$38^2 - 2 \cdot 16 = 1412 \equiv 5$	$38 \cdot (16 - 5) - 63 = 355 \equiv 20$	(5, 20)
Verify: $u_2 \cdot Q = 2 \cdot Q$ (one doubling); $u_1 \cdot G = 4 \cdot G = (25, 17)$ is reused from the key row				
2·(46, 40)	$(3 \cdot 46^2)/(2 \cdot 40) = 50 \cdot 13^{-1} = 50 \cdot 31 \equiv 9$	$9^2 - 2 \cdot 46 = -11 \equiv 56$	$9 \cdot (46 - 56) - 40 = -130 \equiv 4$	(56, 4)
(25, 17) + (56, 4)	$(4 - 17)/(56 - 25) = 54 \cdot 31^{-1} = 54 \cdot 13 \equiv 32$	$32^2 - 25 - 56 = 943 \equiv 5$	$32 \cdot (25 - 5) - 17 = 623 \equiv 20$	(5, 20)

2 · Signing message m_1 = “pay bob 9”

SHA-256(m_1) = 14110b75…; z keeps the leftmost bitlen(79) = 7 bits: $0x14 = 00010100_2 \rightarrow 0001010_2 = z_1 = 10$. r = x(R) mod 79 = 5. k^{-1} mod 79: $14 \cdot 17 = 238 = 3 \cdot 79 + 1 \Rightarrow k^{-1} \equiv 17$. $s = k^{-1}(z_1 + r \cdot d) = 17 \cdot (10 + 5 \cdot 5) = 17 \cdot 35 = 595 \equiv 42 \pmod{79}$. Signature (r, s) = (5, 42). Because $s = 42 > \lfloor 79/2 \rfloor = 39$, this is a high-s signature; its low-s form is (r, 79 − s) = (5, 37).

3 · Verification

$w = s^{-1}$ mod 79: $42 \cdot 32 = 1344 = 17 \cdot 79 + 1 \Rightarrow w = 32$. $u_1 = z_1 \cdot w = 10 \cdot 32 = 320 \equiv 4$, $u_2 = r \cdot w = 5 \cdot 32 = 160 \equiv 2 \pmod{79}$. $X = u_1 G + u_2 Q = (25, 17) + (56, 4) = (5, 20)$ (last table row). x(X) mod 79 = 5 = r ✔. Wrong message: m_2 has z = 3, which gives $u_1 = 17$ and a different X, so verification fails. Low-s twin (5, 37): w = 47, $u_1 = 75$, $u_2 = 77$, X = (5, 47) = −R. x is the same, so it also verifies. That is why (r, s) and (r, n − s) are both valid, and why Bitcoin enforces low-s to remove this malleability.

4 · Nonce reuse: recovering d

The same k = 14 also signs m_2 = “pay alice 5”. SHA-256 = 071251cb…, so $z_2 = 3$. $s_2 = 17 \cdot (3 + 5 \cdot 5) = 17 \cdot 28 \equiv 2$, giving $(r, s_2) = (5, 2)$, which is already low-s. Both signatures share r = 5. After low-s normalisation the published pair is $s_1 = 37$ (flipped: k became −k) and $s_2 = 2$ (not flipped). Subtracting the two equations $s \cdot k = z + r \cdot d$ removes d. The flip is unknown, so try both signs:

(a) same sign: $k = (z_1 - z_2)/(s_1 - s_2) = 7/35$. Euclid gives $35^{-1} \equiv 70$ ($4 \cdot 79 - 9 \cdot 35 = 1$), so $k = 7 \cdot 70 = 490 \equiv 16$. Then $d = (s_1 k - z_1)/r = (37 \cdot 16 - 10) \cdot 5^{-1} = 582 \cdot 16 \equiv 29 \cdot 16 = 464 \equiv 69$. Check: $69 \equiv -10$, so $69 \cdot G = -(2 \cdot Q) = -(56, 4) = (56, 63) \neq Q$. Rejected.

(b) opposite sign: $k = (z_1 - z_2)/(s_1 + s_2) = 7/39$. Since $39 \cdot 2 = 78 \equiv -1$, $39^{-1} \equiv 77$ and $k = 7 \cdot 77 = 539 \equiv 65 = -14$ (the flipped nonce). Then $d = (37 \cdot 65 - 10) \cdot 5^{-1} = 2395 \cdot 16 \equiv 25 \cdot 16 = 400 \equiv 5$, and $5 \cdot G = (46, 40) = Q$ ✔. The private key is recovered.

All four flip combinations (computed by `recover_from_nonce_reuse`; $k' = \pm 14$ is the effective nonce of signature 1): s_1=42, s_2=2 → d = 5 ($k' = 14$) · s_1=42, s_2=77 → d = 5 ($k' = 14$) · s_1=37, s_2=2 → d = 5 ($k' = 65$) · s_1=37, s_2=77 → d = 5 ($k' = 65$). Low-s normalisation only changes the sign of k. It never hides the shared r, so the attack always succeeds after testing at most two candidates against Q.